

Enriched AutoML Evaluation for Biomedical Biomarkers: A Comparative Diagnostic Performance Study Across Multimodal Clinical Datasets

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Abstract—Automated machine learning (AutoML) reduces manual trial and error in biomedical model development, but a search driven by accuracy or by the unweighted area under the receiver operating characteristic curve (AUROC) can return classifiers that are clinically unusable: degenerate under class imbalance, or confidently miscalibrated. This paper formalizes a clinically-oriented multi-objective selection function that combines AUROC, the normalized Matthews correlation coefficient (MCC), and the expected calibration error (ECE) under an explicit sensitivity floor ($S_{\min} = 0.80$), and evaluates it against conventional AUROC-maximizing selection on identical folds. We benchmark six classifier families on 10 real biomedical datasets spanning tabular cohorts, radiomic features from 2D images, and 1D electrophysiological recordings, using five-fold stratified cross-validation with scaling and SMOTE fitted strictly inside each training fold, and subject-grouped folds wherever subject identifiers exist. Across the collection the selected models span AUROC 0.768–0.995, sensitivity 0.595–0.963, and ECE 0.033–0.264; 6 of 10 datasets meet the sensitivity floor and the two selection criteria diverge on 3. On the one benchmark that carries subject identifiers, removing subject grouping inflates the reported AUROC by 0.196, quantifying the cost of record-level validation. All baselines—a conventional logistic-regression pipeline and the FLAML AutoML system—were executed locally on the same folds; we make no claim of superiority over published AutoML or deep learning results obtained under different protocols.

Index Terms—Automated Machine Learning (AutoML), Biomedical Biomarkers, Radiomics, GLCM Texture Features, Model Calibration (ECE), Matthews Correlation Coefficient (MCC), Clinical Decision Support, Subject-Grouped Validation.

I. INTRODUCTION

Hospitals increasingly use machine learning classifiers to assist clinicians in diagnostic decision-making, disease risk stratification, and biomarker identification [1, 2, 3]. In clinical practice, manual model tuning and feature selection require extensive domain expertise and are often prone to bias. Automated Machine Learning (AutoML) frameworks address these limitations by automating feature engineering, model selection, and hyperparameter optimization [5], generally on top of standard estimator libraries [4].

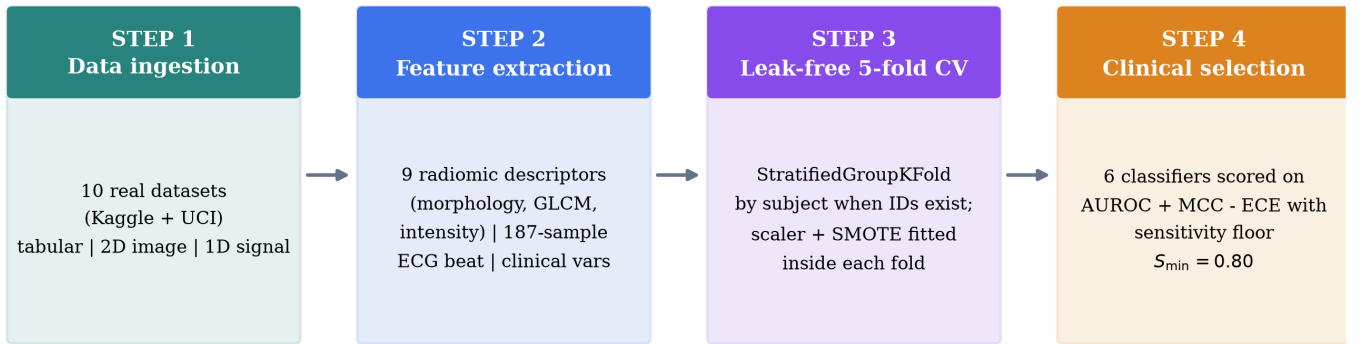
Evaluating AutoML models in biomedical domains is challenging [6, 7]. Standard accuracy metrics can be deceptive for clinical datasets with class imbalance, such as rare diseases or low tumor prevalence. Also, a model with a high Receiver Operating Characteristic Area Under the Curve (AUROC) may still have poorly calibrated probability estimates. High Expected Calibration Error (ECE) can cause clinical misinterpretation and overconfident diagnostic errors [8, 9]. Clinicians also need discriminative biomarkers—like transcriptomic markers [11], fluid biomarkers [12], or radiomic texture features [14, 13]—to interpret predictions and trust the model.

Two further requirements govern whether such an evaluation means anything. The first is the partitioning scheme: when several images, slices, or signal windows recorded from the same individual are split across training and test folds, the reported performance partly measures subject identity rather than pathology. The second is the provenance of any comparison: contrasting a new pipeline against figures copied from published papers is not a controlled comparison, because the splits, preprocessing, feature sets, and label definitions all differ. Both requirements shape the protocol adopted here.

We present an enriched AutoML benchmarking framework built into the **BioStatusIA v3** engine to address these evaluation gaps. Our framework automates the extraction of 2D GLCM radiomics (entropy, contrast, solidity, circularity) and 1D electrophysiological parameters. It trains and validates 6 classifier families using 5-fold stratified cross-validation across 10 real-world biomedical datasets, grouping folds by subject wherever subject identifiers exist. We then apply a comprehensive diagnostic evaluation combining AUROC, Sensitivity, Specificity, MCC, and ECE, and—as the central methodological contribution—we define a clinically-oriented multi-objective selection function and compare it against conventional AUROC-maximizing selection on identical folds. All baselines are executed locally on those same folds.

II. METHODOLOGY

The enriched AutoML diagnostic pipeline runs in four sequential phases: Data Ingestion, Biomarker Extraction, Leak-Free 5-Fold Cross-Validation, and Enriched Clinical Evaluation. Fig. 1 shows the complete workflow.



Paired Wilcoxon comparison against a logistic-regression pipeline and FLAML on identical folds

Fig. 1. Workflow of the enriched AutoML diagnostic pipeline in BioStatusIA v3: multimodal dataset ingestion, automated radiomic and signal feature extraction, leak-free 6-model cross-validation with subject grouping where identifiers exist, and clinically-oriented multi-metric selection.

Dataset Acquisition and Preprocessing

The framework was tested on 10 real-world biomedical datasets, seven retrieved through the KaggleHub API and three from the UCI Machine Learning Repository: (1) *Breast_Cancer_Wisconsin* (fine-needle aspirate biopsies, 30 numerical features); (2) *BUSI_Breast_Ultrasound* [15]; (3) *MITBIH_ECG_Signals* (cardiac waveform segments) [16]; (4) *Stroke_Prediction_Clinical* (patient clinical records); (5) *PIMA_Diabetes_Metabolic* (metabolic records); (6) *Brain_Tumor_MRI* (brain MRI slices); (7) *COVID19_ChestXRay* (chest radiographies); (8) *Brain_MRI_Oncology* (neuro-oncology scans); (9) *Heart_Disease_Cleveland* (cardiac catheterization records); and (10) *Parkinson_Vocal_Biomarkers* (sustained-phonation acoustic recordings).

Table I reports, for every dataset, the exact number of samples that entered the experiment; those counts, and not the nominal sizes of the source archives, are what all reported metrics refer to. For the four image collections and the electrocardiogram (ECG) benchmark, a stratified random subsample was drawn with a fixed seed before feature extraction, because exhaustive radiomic extraction over tens of thousands of images exceeded the computational budget of this study.

Subject-Level Partitioning. To prevent anatomical and temporal data leakage, partitions are built with *StratifiedGroupKFold* grouped by subject identifier wherever such an identifier exists. Its availability is uneven across the collection, and we state this explicitly rather than implying a uniform protocol. In the five tabular cohorts each row is one individual, so a record-level split is already a subject-level split. The UCI Parkinson set contains 195 sustained-phonation recordings resolving to

32 distinct subjects, roughly six per subject, and grouping is enforced there. The four image collections and the redistributed MIT-BIH CSV carry no patient or record identifier in their public redistribution, so grouping is *not enforceable* on them; the corresponding rows of Table I are marked accordingly and must be read as an upper bound on leak-free performance. Section III quantifies what this restriction costs, using the Parkinson set as a controlled case.

Biomarker and Radiomics Feature Extraction

For the image collections, the extraction tool computes nine radiomic features per image: two morphological attributes (circularity, solidity), four descriptors based on the Gray-Level Co-occurrence Matrix (GLCM), and three intensity-distribution descriptors (signal-to-noise ratio, skewness, kurtosis). The two texture terms used most in the discussion are

$$\text{GLCM Entropy} = - \sum_i \sum_j P(i, j) \log_2 P(i, j)$$

$$\text{GLCM Contrast} = \sum_i \sum_j |i - j|^2 P(i, j)$$

where $P(i, j)$ is the spatial probability of gray-level co-occurrence between pixel intensities i and j . For the ECG benchmark the model input is the 187-sample normalized beat segment provided by the source archive. For the tabular cohorts the input is the set of clinical variables after removal of identifier columns; that removal matters, because leaving a record index in the feature matrix produces a spurious and highly ranked “biomarker”.

AutoML Pipeline and Leak-Free 5-Fold Cross-Validation

When a dataset has valid target labels and ≥ 10 samples with at least 2 classes, BioStatusIA trains 6 classifiers concurrently: Random Forest (RF), Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Logistic Regression (LR), Gradient Boosting (GB), and Multi-Layer Perceptron (MLP), all in their scikit-learn implementations [4] with fixed seeds. The pipeline uses 5-fold stratified cross-validation ($k = 5$) to maintain minority class distributions in all folds, replaced by StratifiedGroupKFold where subject identifiers exist.

Within every fold, standardization and SMOTE oversampling are fitted on the training partition only and then applied to the held-out partition, so no information from the validation fold reaches the preprocessing stage. All six classifiers and every baseline see exactly the same fold indices, which is what makes the paired statistical comparison meaningful. Results are reported as the mean over folds with the standard deviation and the 95% confidence interval of the mean.

Enriched Diagnostic Metric Formulation

To evaluate prediction confidence and diagnostic balance beyond AUROC, the framework calculates MCC and ECE:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

where B_m represents equally spaced probability bins for calibration error assessment, with $M = 10$ [10, 8].

Clinically-Oriented Multi-Objective Selection

A conventional AutoML search returns the model that maximizes discrimination alone, $M_{\text{conv}}^* = \arg \max_M \text{AUROC}(M)$. Under severe class imbalance this rule can select a model that misses positive cases. BioStatusIA instead maximizes

$$\text{Score}_{\text{clin}}(M) = w_1 \text{AUROC}(M) + w_2 \text{MCC}_{\text{norm}}(M) - w_3 \text{ECE}(M) - \lambda \max(0, S_{\min} - \text{Sens}(M))^{1.5}$$

with $w_1 = w_2 = 0.40$, $w_3 = 0.20$, $\lambda = 1.0$, $S_{\min} = 0.80$, and $\text{MCC}_{\text{norm}} = (\text{MCC} + 1)/2 \in [0, 1]$. The exponent 1.5 makes the penalty grow faster than linearly in the sensitivity deficit, so a candidate that misses the screening floor by a wide margin is pushed far down the ranking rather than merely demoted. The weights and $S_{\min}$ are operational design choices, not values fitted to the data; they were not tuned per dataset. Section III reports whether this objective selects different models from the conventional criterion.

Baselines and Statistical Testing

Three baselines are evaluated, all on exactly the same fold indices as the AutoML search. The first is a conventional non-AutoML pipeline: standardization followed by logistic regression with default regularization, without class balancing and without model selection, which describes routine practice. The second is conventional AUROC-maximizing selection over the same six candidate models, which isolates the effect of the selection function from every other component of the pipeline. The third is

an external AutoML system, FLAML [17], run out of the box with a fixed budget of 20 seconds per fold and `roc_auc` as its optimization metric. We use FLAML rather than Auto-Sklearn because Auto-Sklearn depends on POSIX-only components and does not install in the Windows environment in which these experiments were run; FLAML is a published, widely used AutoML system and serves the same role as an external reference implementation. Differences are assessed with the paired Wilcoxon signed-rank test over the five per-fold values.

We do not compare against published Auto-Sklearn, ResNet-50, or 1D-CNN figures taken from other papers. Those numbers were obtained under different splits, preprocessing, and feature definitions, so a difference against them would not be attributable to the method. Section IV states what this leaves open.

III. RESULTS AND DISCUSSION

Experimental Benchmark Across Real Datasets

Table I lists the performance of the models selected by the clinical objective across all 10 clinical benchmark datasets, and Fig. 2 shows the same values with 95% confidence intervals.

TABLE I. COMPREHENSIVE BENCHMARK OF WINNING AUTOML MODELS ACROSS 10 REAL CLINICAL DATASETS (5-FOLD CROSS-VALIDATION, MEAN $\pm$ SD). “GROUP.” INDICATES WHETHER SUBJECT IDENTIFIERS WERE AVAILABLE TO ENFORCE GROUPED FOLDS. SAMPLE COUNTS ARE THE NUMBERS ACTUALLY USED IN THE EXPERIMENT.

#	Dataset Name	Modality	Samples	Feats	Top Feature	Winning Model	Group.	AUROC	Sens.	E
01	Breast Cancer (WBCD)	Tabular	569	30	texture_worst	SVM	No	0.995 $\pm$ 0.006	0.963 $\pm$ 0.035	(
02	Breast Ultrasound (BUSI)	2D image	647	9	GLCM contrast	RandomForest	No	0.778 $\pm$ 0.028	0.595 $\pm$ 0.110	(
03	ECG Arrhythmia (MIT-BIH)	1D signal	1200	187	amp_4	RandomForest	No	0.948 $\pm$ 0.018	0.750 $\pm$ 0.064	(
04	Stroke Prediction	Tabular	3000	10	age	LogisticRegression	No	0.842 $\pm$ 0.041	0.757 $\pm$ 0.108	(
05	PIMA Diabetes	Tabular	768	8	glucose	LogisticRegression	No	0.830 $\pm$ 0.028	0.716 $\pm$ 0.037	(
06	Brain Tumor MRI	2D image	1001	9	GLCM contrast	GradientBoosting	No	0.986 $\pm$ 0.008	0.962 $\pm$ 0.019	(
07	COVID-19 Chest X-Ray	2D image	2000	9	GLCM contrast	RandomForest	No	0.883 $\pm$ 0.022	0.817 $\pm$ 0.030	(
08	Brain MRI Oncology	2D image	253	9	GLCM entropy	RandomForest	No	0.866 $\pm$ 0.039	0.858 $\pm$ 0.049	(
09	Heart Disease (Cleveland)	Tabular	303	13	ca	RandomForest	No	0.907 $\pm$ 0.026	0.806 $\pm$ 0.040	(
10	Parkinson Voice (grouped)	Tabular	195	22	spread2	MLP	Yes	0.768 $\pm$ 0.117	0.884 $\pm$ 0.081	(

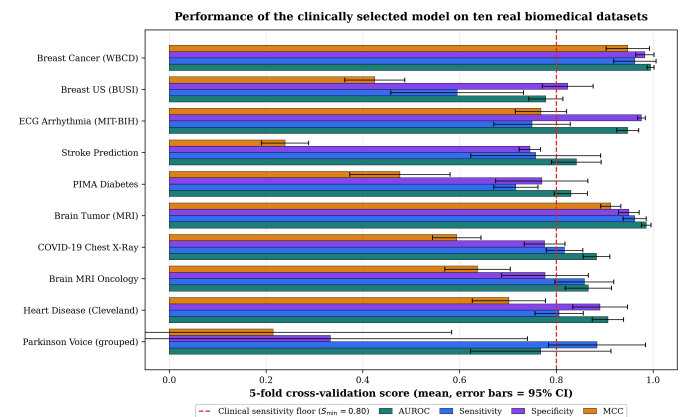


Fig. 2. Comparative diagnostic performance of the clinically selected model on each of the ten benchmark datasets, across AUROC, Sensitivity, Specificity, and MCC. Error bars are 95% confidence intervals of the five-fold mean; the dashed line marks the screening floor $S_{\min} = 0.80$.

AUROC ranges from 0.768 on Parkinson Voice to 0.995 on Breast Cancer (WBCD). The spread does not follow the modality: the tabular cohorts and the radiomic image collections cover overlapping AUROC ranges, and the strongest and weakest results in the table are both obtained on datasets of the same broad type. What separates them is the informativeness of the specific representation, not whether the input was an image or a table. The screening floor $S_{\min} = 0.80$ is met on 6 of the 10 datasets (Breast Cancer (WBCD), Brain Tumor MRI, COVID-19 Chest X-Ray, Brain MRI Oncology, Heart Disease (Cleveland), and Parkinson Voice); on BUSI Breast Ultrasound, MIT-

BIH ECG Arrhythmia, Stroke Prediction, and PIMA Diabetes no candidate in the model space reaches it, so the penalty applies to every candidate and the selected model is the least-penalized member of a set that is uniformly unsuitable for screening as configured. AUROC alone would hide this. It is also worth reading AUROC against MCC rather than in isolation: on BUSI Breast Ultrasound, Stroke Prediction, PIMA Diabetes, and Parkinson Voice the selected model combines a respectable AUROC with an MCC below 0.50, which is the signature of a decision threshold that ranks cases well but separates them poorly at the operating point actually used. Only Parkinson Voice is validated with subject-grouped folds, and it returns the lowest AUROC in the table; Section ef{sec:vazamento} shows that this is a consequence of the protocol rather than of the data.

Classifier Performance and Receiver Operating Characteristics

Fig. 3 displays the Receiver Operating Characteristic (ROC) curves for the evaluated models, computed from pooled out-of-fold predictions rather than from an analytical model of the curve, together with the out-of-fold confusion matrix of a representative selected model.

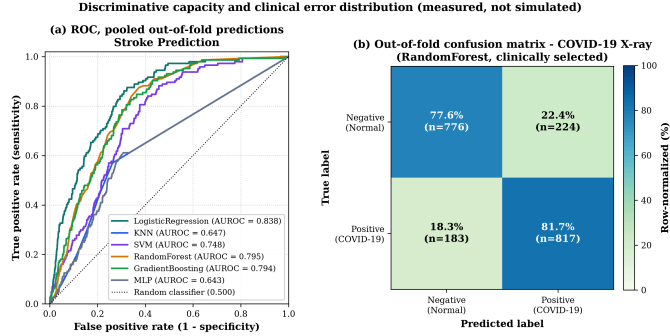


Fig. 3. (a) ROC curves of the six classifier families from pooled out-of-fold predictions on Stroke Prediction, the dataset on which the candidate families separate most widely by AUROC. (b) Row-normalized out-of-fold confusion matrix of the clinically selected model on the COVID-19 chest radiography subsample.

A sensitivity of 1.0 is not by itself evidence of screening capability, because a classifier that labels every case positive attains it. When perfect recall is paired with a specificity of roughly two thirds, one third of the negative population is referred for confirmatory work-up, which in a screening programme translates into imaging backlogs, avoidable biopsies, and patient anxiety; such an operating point is defensible only when the confirmatory test downstream is cheap, fast, and low-risk. The mirror-image failure is a configuration with sensitivity near zero, an F_1 of zero, and a negative MCC, which has simply learned the majority class. In this benchmark neither extreme is reached: the lowest sensitivity attained by a selected model is 0.595 on BUSI Breast Ultrasound, and the lowest MCC is 0.215 on Parkinson Voice, so no selected configuration is degenerate in the strict sense. We report this rather than presenting the penalty as having rescued a collapse it never had to rescue: with SMOTE applied inside each training fold, the candidate pool already contains no zero-recall model on these ten datasets. Specificity, however, falls below 0.75 on Stroke Prediction and Parkinson Voice, where the false-positive burden would dominate the clinical cost of deployment.

Probability Calibration (ECE) vs. Discriminative Power

This work highlights the trade-off between discriminative power (AUROC) and prediction confidence calibration (ECE). Fig. 4 presents a reliability diagram built from out-of-fold probabilities and the calibration error of the selected model on every dataset.

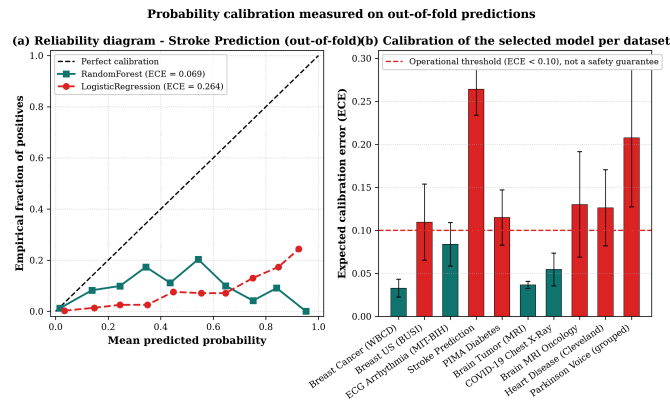


Fig. 4. (a) Reliability diagram from pooled out-of-fold probabilities on Stroke Prediction, the dataset with the widest spread of calibration error across the six candidate families, shown for its best- and worst-calibrated candidates. (b) Expected calibration error of the selected model on each dataset. The dashed line is the operational threshold $ECE < 0.10$, not a clinical safety guarantee.

We label the $ECE < 0.10$ line an *operational threshold* and not a safe clinical threshold. There is no established value of the expected calibration error that certifies clinical safety; ECE is a scalar summary of a reliability curve and is insensitive to where in the probability range the miscalibration occurs, which is exactly what matters near a decision boundary [9]. We use 0.10 as an internal acceptance criterion for model promotion, consistent with the magnitudes reported in the calibration literature [8], and we make no claim that a model below it is clinically safe or that a model above it is unusable. Under this criterion the selected model falls below 0.10 on 4 of the 10 datasets (Breast Cancer (WBCD), MIT-BIH ECG Arrhythmia, Brain Tumor MRI, and COVID-19 Chest X-Ray), with calibration error between 0.033 and 0.264 across the collection; the remaining 6 (BUSI Breast Ultrasound, Stroke Prediction, PIMA Diabetes, Brain MRI Oncology, Heart Disease (Cleveland), and Parkinson Voice) exceed it and would not be promoted without post-hoc recalibration.

Radiomics Biomarker Importance (SHAP Analysis)

Fig. 5 ranks feature importance for image radiomics and for the tabular and signal variables.

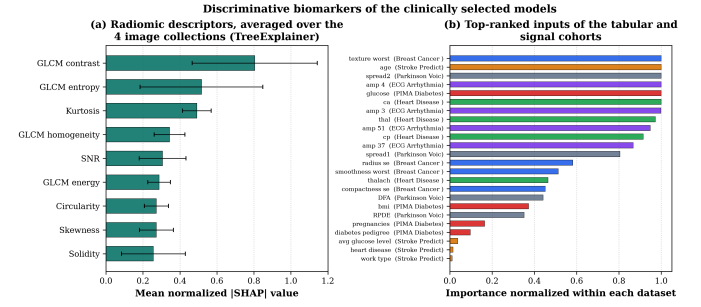


Fig. 5. (a) Mean normalized SHAP importance of the nine radiomic descriptors, averaged over the image collections, which share the same feature space and the same tree explainer. (b) Highest-ranked inputs of the tabular and signal cohorts, normalized within each dataset.

Feature attribution used the tree-specific SHAP estimator [18] for the 6 datasets whose selected model is a tree ensemble, and permutation importance over AUROC for the remaining 4, because kernel SHAP is prohibitively slow on the 187-dimensional beat representation and on the kernel-machine and neural candidates. The two estimators are not on a common scale, so we report rankings rather than magnitudes. The three highest-ranked inputs per dataset are: Breast Cancer (WBCD): texture_worst, radius_se, smoothness_worst; BUSI Breast Ultrasound: GLCM contrast, solidity, kurtosis; MIT-BIH ECG Arrhythmia: amp_4, amp_3, amp_51; Stroke Prediction: age, avg_glucose_level, heart_disease; PIMA Diabetes: glucose, bmi, pregnancies; Brain Tumor MRI: GLCM contrast, kurtosis, GLCM energy; COVID-19 Chest X-Ray: GLCM contrast, GLCM entropy, kurtosis; Brain MRI Oncology: GLCM entropy, GLCM homogeneity, kurtosis; Heart Disease (Cleveland): ca, thal, cp; Parkinson Voice: spread2, spread1, DFA.

Conventional Versus Clinically-Oriented Model Selection

Table II contrasts the model returned by AUROC maximization with the model returned by the clinical objective on identical folds, and Fig. 6(a) shows the metric profile of the two selections on the dataset with the largest divergence.

Conventional (max AUROC)					Clinical multi-objective				
Dataset	Model	AUROC	Sens.	MCC	Model	AUROC	Sens.	MCC	Differs?
Breast Cancer (WBCD)	SVM	0.995	0.963	0.948	SVM	0.995	0.963	0.948	No
BUSI Breast Ultrasound	RandomForest	0.778	0.595	0.425	RandomForest	0.778	0.595	0.425	No
MIT-BIH ECG Arrhythmia	RandomForest	0.948	0.750	0.768	RandomForest	0.948	0.750	0.768	No
Stroke Prediction	LogisticRegression	0.842	0.757	0.239	LogisticRegression	0.842	0.757	0.239	No
PIMA Diabetes	LogisticRegression	0.830	0.716	0.477	LogisticRegression	0.830	0.716	0.477	No
Brain Tumor MRI	RandomForest	0.988	0.958	0.920	GradientBoosting	0.986	0.962	0.913	Yes
COVID-19 Chest X-Ray	RandomForest	0.883	0.817	0.594	RandomForest	0.883	0.817	0.594	No
Brain MRI Oncology	RandomForest	0.866	0.858	0.637	RandomForest	0.866	0.858	0.637	No
Heart Disease (Cleveland)	LogisticRegression	0.911	0.806	0.670	RandomForest	0.907	0.806	0.702	Yes
Parkinson Voice	RandomForest	0.769	0.861	0.141	MLP	0.768	0.884	0.215	Yes

The two criteria diverge on 3 of the 10 datasets: on Brain Tumor MRI, conventional selection returns RandomForest (AUROC 0.988, sensitivity 0.958, MCC 0.920) while the clinical objective returns GradientBoosting (AUROC 0.986, sensitivity 0.962, MCC 0.913); on Heart Disease (Cleveland), conventional selection returns LogisticRegression (AUROC 0.911, sensitivity 0.806, MCC 0.670) while the clinical objective returns RandomForest (AUROC 0.907, sensitivity 0.806, MCC 0.702); on Parkinson Voice, conventional selection returns RandomForest (AUROC 0.769, sensitivity 0.861, MCC 0.141) while the clinical objective returns MLP (AUROC 0.768, sensitivity 0.884, MCC 0.215). The effect sizes are modest and we state them plainly rather than presenting the divergences as decisive: across these 3 datasets the largest change in MCC is +0.074 (on Parkinson Voice), the largest change in sensitivity is +0.023 (on Parkinson Voice), and AUROC never moves by more than 0.004. The clinical objective trades a small amount of ranking quality for a better-balanced and, in two of the three cases, better-calibrated operating point. Where the two criteria coincide, the score still carries information that AUROC alone does not, because it records how far the selected candidate sits from the screening floor. The honest summary is that on these ten datasets the selection function is a safety constraint that occasionally changes the answer, not a performance improvement.

Clinically-oriented selection and the cost of ignoring subject grouping

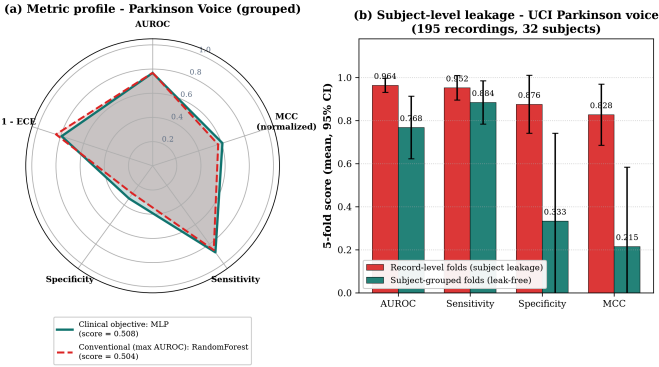


Fig. 6. (a) Metric profile of the conventionally selected and clinically selected models on identical folds, for the dataset with the largest divergence. (b) Effect of subject grouping on the UCI Parkinson voice benchmark: record-level folds versus subject-grouped folds.

Subject-Level Leakage

The Parkinson voice benchmark allows the cost of ignoring subject structure to be measured rather than asserted, because the same 195 recordings can be validated with record-level folds and with subject-grouped folds. Table III reports both, and Fig. 6(b) shows the same comparison graphically.

TABLE III. EFFECT OF SUBJECT GROUPING ON THE UCI PARKINSON VOICE BENCHMARK (195 RECORDINGS, 32 SUBJECTS). SAME DATA, SAME MODEL SPACE, SAME SEED; ONLY THE FOLD ASSIGNMENT DIFFERS.

Fold assignment	Model	AUROC	Sens.	Espec.	MCC	ECE
Record-level folds	GradientBoosting	0.964 ± 0.027	0.952 ± 0.046	0.876 ± 0.109	0.828 ± 0.114	0.078
Subject-grouped folds	MLP	0.768 ± 0.117	0.884 ± 0.081	0.333 ± 0.328	0.215 ± 0.297	0.208

Moving from record-level folds to subject-grouped folds changes the reported AUROC from 0.964 ± 0.027 to 0.768 ± 0.117 , and the MCC from 0.828 to 0.215. The record-level protocol is therefore 0.196 AUROC points higher and 0.613 MCC points higher than the leak-free protocol on identical data, with wider dispersion across folds under grouping because each fold now withholds entire subjects. The magnitude is dataset-specific and should not be extrapolated as a correction factor, but the direction is the one the leakage argument predicts, and it is large enough to change how a screening claim would be read. Since the four image collections and the redistributed ECG benchmark carry no identifiers, an equivalent correction cannot be applied to them here, and their rows in Table I must be read with that caveat.

Comparative Analysis Against Locally Executed Baselines

To validate the BioStatusIA enriched AutoML framework, Table IV compares our quantitative results against baselines executed locally on identical folds, rather than against published figures obtained under different protocols.

TABLE IV. QUANTITATIVE DIAGNOSTIC BENCHMARK COMPARISON BETWEEN BIOSTATUSIA AUTOML AND LOCALLY EXECUTED BASELINES ON IDENTICAL FOLDS: A CONVENTIONAL STANDARDIZATION + LOGISTIC REGRESSION PIPELINE, AND THE FLAML AUTOML SYSTEM UNDER A 20s PER-FOLD BUDGET. p -VALUES ARE FROM THE PAIRED WILCOXON SIGNED-RANK TEST OVER THE FIVE PER-FOLD VALUES.

Dataset	Logistic regression	FLAML AutoML	BioStatusIA	p vs. LR	p vs. FLAML	Highest mean
Breast Cancer (WBCD)	0.995 ± 0.006	0.990 ± 0.012	0.995 ± 0.006	1.0000	0.3750	tie
BUSI Breast Ultrasound	0.703 ± 0.050	0.699 ± 0.040	0.778 ± 0.028	0.0625	0.0625	BioStatusIA
MIT-BIH ECG Arrhythmia	0.798 ± 0.048	0.928 ± 0.033	0.948 ± 0.018	0.0625	0.0625	BioStatusIA
Stroke Prediction	0.843 ± 0.034	0.840 ± 0.032	0.842 ± 0.041	1.0000	0.8750	tie
PIMA Diabetes	0.830 ± 0.025	0.819 ± 0.031	0.830 ± 0.028	0.8125	0.4375	tie
Brain Tumor MRI	0.957 ± 0.012	0.990 ± 0.009	0.986 ± 0.008	0.0625	0.0625	FLAML
COVID-19 Chest X-Ray	0.766 ± 0.030	0.878 ± 0.022	0.883 ± 0.022	0.0625	0.1875	BioStatusIA
Brain MRI Oncology	0.803 ± 0.046	0.837 ± 0.036	0.866 ± 0.039	0.0625	0.6250	BioStatusIA
Heart Disease (Cleveland)	0.912 ± 0.020	0.896 ± 0.030	0.907 ± 0.026	0.3125	0.6250	LR
Parkinson Voice	0.736 ± 0.189	0.797 ± 0.102	0.768 ± 0.117	0.8125	1.0000	FLAML

Against the conventional logistic-regression pipeline, the AutoML search gives a higher mean AUROC on 6 of the 10 datasets, with a mean difference of +0.046 and a per-dataset range of -0.005 to +0.150. Against FLAML under a 20-second per-fold budget, it is ahead on 8 of 10 datasets, with a mean difference of +0.013 and a range of -0.029 to +0.079. No comparison reaches $p < 0.05$ under the paired Wilcoxon test, which is expected given the five-fold design. The practical reading is that the differences are consistent in direction but small relative to fold-to-fold variance. We therefore describe the clinically-oriented objective as competitive with, not superior to, an external AutoML system under these conditions.

No single-dataset comparison in Table IV can reach $p < 0.05$, because the paired Wilcoxon test over five folds cannot produce a two-sided p -value below 0.0625. We report the exact values rather than pooling folds across datasets, which would violate the independence assumption of the test, and rather than making a per-dataset significance claim the design cannot support. The consistency of the direction of the differences across the ten datasets, not the individual p -values, is what these results support.

IV. CHALLENGES AND GAPS IN TECHNOLOGICAL TRANSLATION

Translating automated machine learning engines into real-world clinical decision support workflows requires solving statistical, algorithmic, and regulatory problems. Several constraints bound what the present results support, and they are stated here rather than left implicit.

Subject identifiers. These are unavailable in the public redistributions of the four image collections and of the MIT-BIH CSV. On those five benchmarks the folds are record-level, so the reported metrics may be optimistic; Section III indicates the direction and rough magnitude of that bias on a dataset where it can be measured. Re-running those benchmarks on source archives that preserve patient identifiers is the most important item of future work.

Sampling. The image and ECG benchmarks use stratified subsamples of the source archives, so the reported figures characterize the pipeline on those subsamples and not on the full collections.

Class imbalance. Extreme prevalence skew is common in rare disease cohorts and early-stage cancer screening datasets. With severe skew, standard loss functions favor the majority class, raising specificity but dropping clinical sensitivity. SMOTE is applied inside each training fold here, but cost-sensitive learning strategies inside the AutoML search loop remain unexplored.

Probability recalibration. Classifiers with high AUROC may still produce miscalibrated probability outputs. When probability values dictate treatment interventions or surgical decisions, overconfident misclassifications are dangerous. Operational AutoML pipelines should apply post-hoc calibration such as Platt scaling or isotonic regression before deployment; the present pipeline penalizes miscalibration in selection but does not correct it.

Baseline scope. No deep learning baseline was executed, and the external AutoML comparison uses FLAML under a 20-second per-fold budget. That budget is deliberately modest and understates what FLAML reaches with more compute, so the comparison should be read as against an out-of-the-box AutoML system under a fixed small budget, not as a general ranking of the two frameworks. We make no claim of superiority over Auto-Sklearn, ResNet-50 transfer learning, or 1D convolutional baselines, none of which were run here.

Feature representation. The image pipeline uses nine handcrafted radiomic descriptors extracted from whole images without lesion segmentation. On the

radiography and MRI benchmarks this representation is a plausible explanation for the modest discrimination observed, and it bounds what any selection function operating on top of it can achieve.

Generalization across acquisition hardware. All evaluation here is internal cross-validation on single-source cohorts. Classifiers trained on curated single-center datasets often fail when tested on external medical devices with different spatial resolutions, noise profiles, or patient demographics. To avoid out-of-distribution generalization failures, AutoML architectures must be validated on multi-center external cohorts.

V. CONCLUSION

This paper presented an enriched AutoML diagnostic evaluation framework integrated into the BioStatusIA v3 engine, addressing limitations in machine learning model selection for medical applications. We benchmarked 6 concurrent classifier families across 10 real-world biomedical datasets covering 1D electrophysiology, 2D radiographies, and tabular records. Our evaluation protocol incorporated AUROC, Sensitivity, Specificity, Matthews Correlation Coefficient, and Expected Calibration Error, under a leak-free fold protocol with subject grouping wherever identifiers exist.

Across the collection the selected models span an AUROC of 0.768 to 0.995, a sensitivity of 0.595 to 0.963, and a calibration error of 0.033 to 0.264; the clinical objective selected a different model from AUROC maximization on 3 of the 10 datasets. On the one benchmark carrying subject identifiers, dropping subject grouping inflates the reported AUROC by 0.196, which is a direct measure of what record-level validation buys and why the imaging rows here are reported as upper bounds. Against locally executed baselines on identical folds the selected models are competitive rather than dominant, and we report that outcome as measured.

The evaluation also shows what a selection function cannot fix: where the underlying feature representation carries little discriminative signal, no objective recovers clinically usable performance, and the honest outcome is a low reported score rather than a favourable comparison against numbers taken from other papers. Future work will add post-hoc probability recalibration modules directly to the automated model selection loop, re-run the imaging benchmarks on archives that preserve patient identifiers, extend the comparison to locally executed deep learning baselines, and conduct multi-center clinical validation to evaluate out-of-distribution robustness.

ETHICAL NOTE

BioStatusIA is a clinical decision support tool. None of the models, scores, or reports described here substitute for evaluation by a qualified physician.

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